Reminder of basic algebra functions for calculus.

With calculus, the derivative looks at the rate of change of a function at any given point.

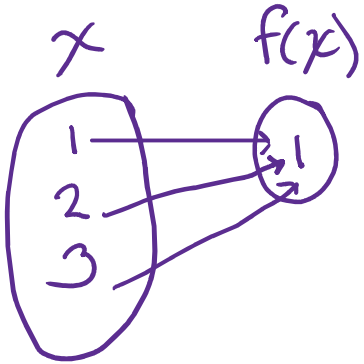
The integral looks at the area under a function, or the cumulative output of a function between two points.

We will start by reviewing basic functions.

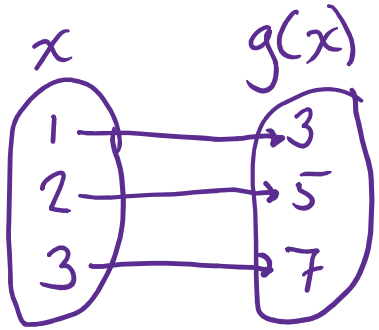
What is a function?

A function is a relation between two sets, an input set and an output set that has a unique output for any given input. We can best see this with a mapping or a graph.

Mapping:



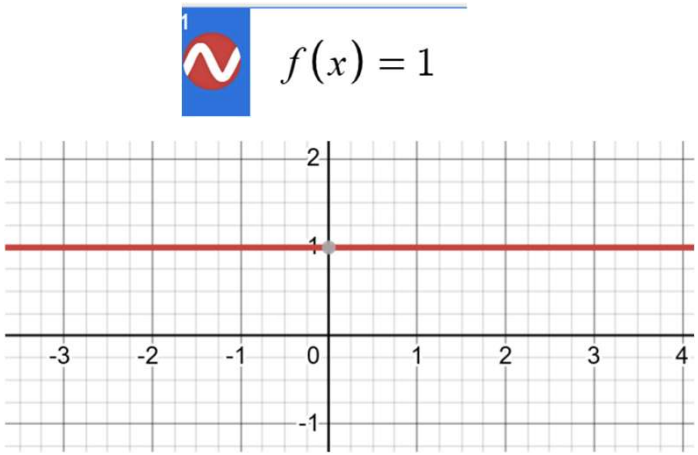
$f(x) = 1$

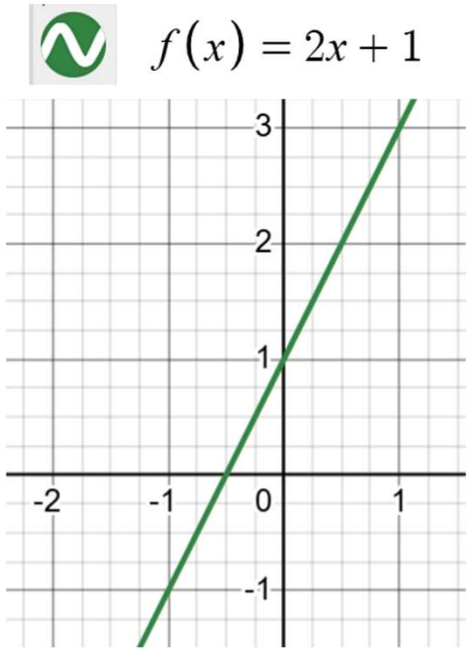


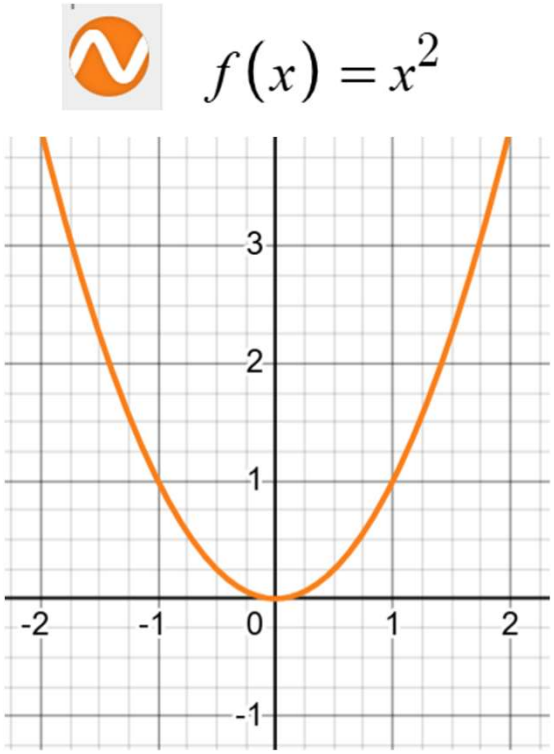
$g(x) = 2x + 1$

We can graph functions with numbers on a coordinate axis system.

These and all similar graphs are done in Desmos. [Desmos | Graphing Calculator]







Function notation uses f, g, and h most commonly to refer to the function and then the input is in parentheses. This equals a formula for finding any given point on the function for that input.

function name

input

output

$f(x) = 1$

$f(a) = 1$

$f(2) = 1$

$g(x) = 2x+1$

$g(a) = 2a+1$

$g(2) = 2 \cdot 2 + 1$

$h(x) = x^2$

$h(a) = a^2$

$h(2) = 2^2$

A polynomial is a function that is a sum of one or more terms, each of which is a constant multiplied by the input variable raised to a non-negative integer power.

When there is one term, it is a monomial.

$3x^2$ $\quad$ $2x^4$

Binomials have two terms.

$3x^2 + 2x$ $\quad$ $2x + 1$

Trinomials have three terms.

$2x^2 + 4x + 2$

Any number of terms is a polynomial.

$4x^4 + 3x^3 + 2x^2 + x + 1$

We write them in order from the highest power of the variable to the lowest. The number each term is multiplied by is called the coefficient.

$$2x^3 + 4x^2 + 5x$$

The first term with the highest power is called the leading term and its coefficient is the leading coefficient.

We can write polynomials with series notation:

$$a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0 x^0$$

Since x to the zero is one, the last becomes the constant term.

We will be talking about the basic or parent functions for each type of function.

Constant function $f(x) = c$

Linear function $f(x) = x$

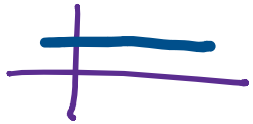
Quadratic function $f(x) = x^2$

Cubic function $f(x) = x^3$

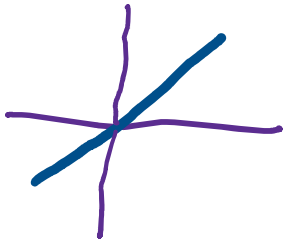
General polynomial function $f(x) = x^n$

Graphs of the parent functions:

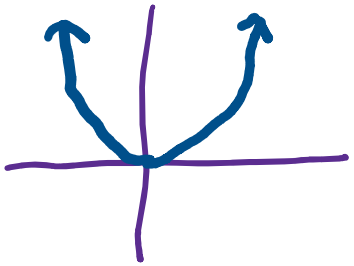
Constant function $f(x) = c$



Linear function $f(x) = x$



Quadratic function $f(x) = x^2$



If you need more review, you can go to those units in the Crainix Understanding Advanced Maths series.